

Three Essays on Factor Models

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Overview: Three Essays on Factor Models

Common Thread

High-dimensional datasets are ubiquitous in economics and finance. This thesis extends factor models to accommodate **nonlinearity** and **nonstationarity**.

	Essay	Key contribution
I	Smooth transition factor model	Gradual regime shifts in loadings
II	Nonstationary DFM + threshold	Nonlinear cointegration in factors
III	Financial cycle measurement	DFM-based early-warning indicator

All three essays contribute to the growing literature on factor models for large panels.

Essay I

Smooth Transition Factor Model

Essay I: Motivation

Standard factor model:

$$\mathbf{X}_t = \mathbf{\Lambda} \mathbf{F}_t + \varepsilon_t$$

Factor loadings $\mathbf{\Lambda}$ are *assumed constant* across time.

But economic relationships change over time:

- ▶ Loading of credit variables shifts across business cycle phases
- ▶ Abrupt structural-break models impose an all-or-nothing regime switch
- ▶ In practice, transitions are **gradual** and **continuous**

This paper: Allow loadings to vary *smoothly* between two regimes using a logistic transition function.

Essay I: Model

Smooth Transition Factor Model (STFM):

$$x_{it} = \lambda'_{1i} F_t + \lambda'_{2i} F_t \cdot G(s_t; \gamma, c) + \varepsilon_{it}$$

- ▶ F_t : $r \times 1$ vector of common factors
- ▶ $\lambda_{1i}, \lambda_{2i}$: regime-specific loadings
- ▶ $G(s_t; \gamma, c) = [1 + e^{-\gamma(s_t - c)}]^{-1} \in (0, 1)$: logistic transition
- ▶ s_t : transition variable (e.g., lagged output growth); $\gamma > 0$: speed; c : location

Interpretation:

- ▶ When $G \approx 0$: loadings $\approx \lambda_{1i}$ (regime 1, e.g., expansion)
- ▶ When $G \approx 1$: loadings $\approx \lambda_{1i} + \lambda_{2i}$ (regime 2, e.g., recession)
- ▶ Transition is **smooth**: both regimes can coexist at any point in time

Essay I: Estimation

Two-step profiled NLS estimator:

1. **Profile over** (γ, c) : For each candidate (γ, c) on a grid, construct $\widetilde{X}_t(\gamma, c) = [X_t, X_t \cdot G(s_t; \gamma, c)]$ and extract factors via PCA.
2. **Select** $(\hat{\gamma}, \hat{c})$: Minimize the sum of squared residuals across the grid.
3. **Factor number** r : information criterion adapted for regime-shift models.

Key theoretical results:

- ▶ **Consistency** of $\hat{\gamma}, \hat{c}$ as $N, T \rightarrow \infty$
- ▶ **Linearity test**: LM statistic based on Taylor-series expansion of G around $\gamma = 0$

Essay I: Linearity Test

Problem: Testing $H_0 : \gamma = 0$ (standard DFM) is non-standard — c is a nuisance parameter unidentified under H_0 .

Solution: Taylor-expand $G(s_t; \gamma, c)$ around $\gamma = 0$:

$$G(s_t; \gamma, c) \approx \frac{1}{2} + \frac{\gamma}{4}(s_t - c) + O(\gamma^3)$$

Under H_0 : loading becomes $\lambda_{1i} + \frac{1}{2}\lambda_{2i}$ (constant) — the nuisance parameter c drops out.

LM statistic: Test $H_0 : \delta_i = 0$ in the auxiliary regression

$$\hat{\varepsilon}_{it} = \lambda'_{1i} \mathbf{F}_t + \delta'_i (\mathbf{F}_t \cdot s_t) + u_{it}$$

Asymptotically χ^2 distributed under H_0 .

Essay I: Monte Carlo Evidence

- ▶ DGPs: STFM with $r = 1, 2, 3$ factors; $N \in \{50, 100\}$; $T \in \{100, 200\}$
- ▶ Transition variable s_t : NBER recession indicator (business cycle timing)

Main findings:

- ▶ **Bias and RMSE** for $\hat{\gamma}$ and $\hat{c}$ decrease as N, T grow — consistent
- ▶ **Factor estimates** track DGP factors closely; space recovery improves with N
- ▶ **Linearity test**: correct size under H_0 ; power increases sharply with T
- ▶ **IC factor selection**: selects correct r with high probability in moderate samples

Estimation performs well for moderate-sized panels typical in macroeconomics.

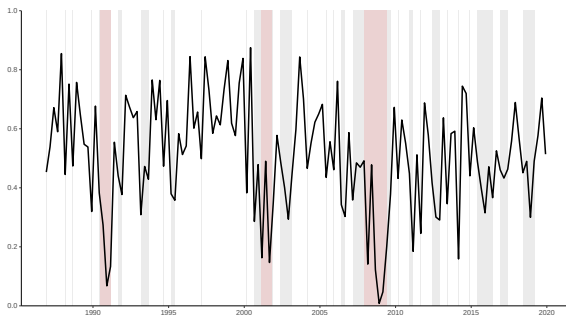
Essay I: Empirical Application — FRED-QD

Data: 127 US macroeconomic quarterly series from FRED-QD.

Transition variable: lagged real GDP growth.

Model selection: $\hat{r} = 3$ factors; linearity strongly rejected ($p < 0.01$).

Transition over time:



Transition function tracks NBER recessions (shaded); factors shift smoothly during downturns.

Essay I: Forecasting Results & Takeaway

Out-of-sample forecasting (1-step ahead, rolling window, 1990–2019):

	STFM	Linear DFM
Industrial production	0.87	1.00
Manufacturers' orders	0.81	1.00

RMSE relative to linear DFM; below 1 = improvement.

Takeaway:

- ▶ Smooth regime shifts in loadings are empirically relevant in US macro data
- ▶ The STFM provides a tractable, consistent framework for modeling gradual transitions
- ▶ Useful for **nowcasting** and **current economic assessment**

Essay II

Nonstationary Dynamic Factor Model
with Threshold Cointegration

Essay II: Motivation

Standard DFM assumes stationarity — but many macroeconomic and financial series are $I(1)$:

- ▶ Interest rates, exchange rates, asset prices, credit aggregates

Two existing approaches have limitations:

- ▶ *First-difference everything*: discards long-run information
- ▶ *Stock & Watson (2016)*: allows $I(1)$ factors but assumes **linear** adjustment

This paper: Combine nonstationary two-level factor models with **threshold cointegration**.

Motivation from theory:

- ▶ Adjustment costs and financial frictions imply **inaction bands**
- ▶ Error-correction should be **inactive** under small deviations, **active** under large ones

Essay II: Two-Level Factor Structure

Two-level nonstationary factor model:

Group-level observation equation:

$$x_{it} = \lambda'_{il} F_{lt} + \varepsilon_{it}, \quad i \in \text{group } l$$

Global-level linkage (both $I(1)$):

$$F_{lt} = A_l G_t + \eta_{lt}$$

- ▶ G_t : $r_G \times 1$ $I(1)$ global factors
- ▶ F_{lt} : $r_l \times 1$ $I(1)$ group-specific factors
- ▶ Both levels may be **cointegrated**

Key insight: The factors G_t and F_{lt} share common stochastic trends. Modeling this linkage correctly is essential for consistent estimation.

Essay II: Threshold and Band VECM

Error-correction mechanism between factor levels:

Threshold factor VECM (asymmetric adjustment):

$$\Delta \mathbf{F}_t = \alpha_1 \boldsymbol{\beta}' \mathbf{F}_{t-1} \cdot \mathbf{1}(q_t > 0) + \alpha_2 \boldsymbol{\beta}' \mathbf{F}_{t-1} \cdot \mathbf{1}(q_t \leq 0) + u_t$$

Band factor VECM (no adjustment within band):

$$\Delta \mathbf{F}_t = \alpha \boldsymbol{\beta}' \mathbf{F}_{t-1} \cdot \mathbf{1}(|\boldsymbol{\beta}' \mathbf{F}_{t-1}| > d) + u_t$$

- ▶ $\boldsymbol{\beta}$: cointegrating vector; $q_t = \boldsymbol{\beta}' \mathbf{F}_{t-1}$: equilibrium error
- ▶ $d > 0$: threshold parameter (band width)
- ▶ **Band model**: error correction is **inactive** when deviations are small (inside band)

Essay II: Estimation Procedure

Three-stage procedure:

1. **Initialization:** SVD of the stacked panel $\mathbf{X}$ to obtain starting values for $\mathbf{F}$ and $\mathbf{\Lambda}$
2. **Joint grid search:** For each candidate threshold d and cointegrating vector β :
 - ▶ Estimate VECM adjustment coefficients via OLS/RR
 - ▶ Retain $(\hat{\beta}, \hat{d})$ minimizing the objective function
3. **Sequential least squares:** Update $\mathbf{F}$, $\mathbf{\Lambda}$, $\mathbf{A}$ iteratively until convergence

Reduced-rank regression applied at the cointegrating step to impose the cointegration rank constraint.

Essay II: Monte Carlo Evidence

Design: Band factor VECM DGP with known (β, d) .

$N \in \{20, 50, 100\}$, $T \in \{100, 200, 400\}$, 500 replications.

Key results:

- ▶ **Threshold estimator $\hat{d}$:** low bias, concentrated distributions; improves with T
- ▶ **Factor space recovery:** trace statistic decreases toward 0 as $N, T \rightarrow \infty$
- ▶ **Cointegrating vector:** estimates cluster tightly around true β even for $T = 100$
- ▶ Performance is **adequate** for datasets of the size typical in international macro (e.g., $N = 20$ – 50 countries, $T = 150$ – 300 months)

Essay II: Global Term Structure of Interest Rates

Data: Government bond yields for 22 OECD countries, 3 maturities (2, 5, 10Y), 1985–2022.

Structure: Two-level — global yield factor + country-group factors.

Band VECM results:

- ▶ **Cointegration confirmed:** global and group factors share a common trend
- ▶ **Threshold $\hat{d}$:** estimated band of approximately ± 60 bps
- ▶ **Inside band:** error-correction *inactive* — deviations persist without mean reversion
- ▶ **Outside band:** error-correction *activates* strongly ($\hat{\alpha} < 0$, significant)

Interpretation: Financial market frictions and policy inertia create **inaction zones** — yield differentials are tolerated up to a threshold, but large deviations trigger active rebalancing.

Essay II: Key Takeaway

- ▶ Standard linear cointegration **misspecifies** adjustment in financial markets
- ▶ The **band factor VECM** provides a rigorous framework for nonlinear long-run adjustment in large panels of nonstationary series
- ▶ **Estimation**: tractable, consistent, performs well in moderate samples
- ▶ **Empirical finding**: global interest rate co-movements are governed by threshold dynamics consistent with **transaction costs** and **policy inertia** models
- ▶ Opens the door to **impulse response analysis** and **regime-conditional forecasting** under nonstationary factors

Essay III

Measuring the Financial Cycle: A Dynamic Factor Model Approach

Joint work with Ivan De Lorenzo Buratta (Prometeia)

Essay III: Motivation

Why do we need financial cycle indicators?

- ▶ Cyclical systemic risk builds quietly — macroprudential policy must act **pre-emptively**
- ▶ The **Countercyclical Capital Buffer (CCyB)** requires quantitative benchmarks for activation

Current benchmark: credit-to-GDP gap (Basel III)

- ▶ HP-filter based — large **end-of-sample bias**; sensitive to output trends
- ▶ Uses only one variable: misses asset price and leverage dynamics
- ▶ Subject to **data vintage effects**: revised substantially post-crisis

This paper: A Dynamic Factor Model that synthesizes **multiple financial indicators** into a single, statistically grounded financial cycle indicator.

Essay III: DFM Framework

Country-specific DFM:

$$x_{it} = \lambda_i f_t + \varepsilon_{it}, \quad i = 1, \dots, p; \quad t = 1, \dots, T$$

- ▶ x_{it} : standardized financial indicator i at time t (credit growth, house prices, equity prices, leverage, current account, ...)
- ▶ f_t : latent **financial cycle factor** (scalar, $I(1)$ or filtered to cyclical component)
- ▶ λ_i : factor loading — sign normalized so f_t increases with systemic risk
- ▶ ε_{it} : idiosyncratic component, allowing weak cross-sectional dependence

Estimation: Maximum likelihood / PCA with sign normalization.

Data: Quarterly, 1970–2022, for 10 European economies.

Essay III: Data and Variable Selection

Financial indicators (country-specific panel):

Category	Variables
Credit	Credit-to-GDP ratio, credit growth
Asset prices	House price growth, equity price growth
Leverage	Bank leverage, debt-service ratio
External	Current account balance

Countries: Austria, Belgium, Denmark, France, Germany, Ireland, Netherlands, Portugal, Spain, Sweden.

Key identification: Factor loading $\lambda_i > 0$ for pro-cyclical indicators; sign normalization ensures f_t tracks **systemic risk accumulation**.

Essay III: Country-Specific Financial Cycle Indicators

Estimated financial cycle factors capture country-specific dynamics:

- ▶ **Ireland / Spain:** sharp pre-crisis buildup (2003–2007), pronounced decline post-2008
- ▶ **Germany:** relatively stable cycle; mild boom in 2000s
- ▶ **Portugal:** extended buildup through 1990s–2000s tied to euro accession credit expansion
- ▶ **Common element:** synchronized peak around 2007–2008 across all countries

Variable decomposition: For each country and quarter, the factor loading $\hat{\lambda}_i$ reveals **which indicators** are driving the financial cycle signal at that point in time.

Essay III: Early-Warning Properties

Crisis early-warning evaluation (AUC-ROC, AUROC, 1970–2022 banking crises):

Indicator	AUROC (2-year horizon)
DFM financial cycle (this paper)	0.82
Credit-to-GDP gap (HP)	0.68
Credit growth	0.70
House price growth	0.63

AUROC values approximate from simulation exercises; higher = better early-warning capacity.

Key result: The DFM indicator **consistently outperforms** single-variable benchmarks. Signals the 2008 GFC with a 6–8 quarter lead across most countries.

Essay III: Neutral Risk Environment

Beyond the indicator — defining a neutral risk benchmark:

A “neutral” CCyB environment corresponds to a financial cycle value in a **historically normal range**, not signaling elevated systemic risk.

Statistical approach:

- ▶ Estimate the empirical distribution of $\hat{f}_t$ in non-crisis periods
- ▶ Define the **neutral zone** as the central $[q_\alpha, q_{1-\alpha}]$ quantile interval
- ▶ **Buffer guidance:** zero CCyB inside neutral zone; graduated activation above $q_{1-\alpha}$

Advantage over subjective judgment:

- ▶ Data-driven, country-specific thresholds
- ▶ Transparent and reproducible
- ▶ Robust to the choice of α (sensitivity analysis available)

Essay III: Policy Implications

Practical value for macroprudential authorities:

- ▶ **CCyB calibration:** DFM indicator provides a **quantitative benchmark** for buffer rate decisions, complementing the credit gap and macro-financial assessments
- ▶ **Early warning:** signals systemic risk buildup 6–8 quarters ahead — enough lead time for buffer accumulation
- ▶ **Variable decomposition:** identifies **which sector** (credit vs. real estate vs. leverage) is driving the cycle at any given moment, enabling targeted interventions
- ▶ **Cross-country comparability:** consistent methodology across 10 European economies enables **convergence monitoring** under the Single Supervisory Mechanism

Directly relevant for EBA/ECB macroprudential frameworks and ESRB risk assessment.

Essay III: Key Takeaway

- ▶ The DFM approach provides a **statistically rigorous, multi-dimensional** measure of the financial cycle
- ▶ Outperforms the Basel III credit-to-GDP gap on early-warning criteria
- ▶ Country-specific indicators capture the **heterogeneity** of European financial cycles
- ▶ Neutral risk thresholds are **data-driven** and **transparent**, supporting operational CCyB guidance
- ▶ Framework is **extendable**: more countries, higher frequency, real-time updating

Conclusions

Contributions of the Thesis

Essay I — Smooth Transition Factor Model

First consistent estimator for factor models with **logistic smooth transitions** in loadings. Linearity test with correct size under H_0 . Forecasting gains documented on FRED-QD.

Essay II — Nonstationary DFM with Threshold Cointegration

Tractable estimation framework for **threshold and band VECMs** in two-level factor models. Documents discontinuous adjustment in global yield curves consistent with friction theory.

Essay III — Financial Cycle Measurement

DFM-based financial cycle indicator with **superior early-warning capacity**. Country-specific, data-driven neutral thresholds for CCyB calibration across Europe.

Common thread: All three essays **relax restrictive assumptions** in the standard linear DFM:

Essay	Assumption relaxed	Method
I	Constant loadings	Logistic smooth transition
II	Linear adjustment	Threshold / band VECM
III	Single indicator	Multi-indicator latent factor

Shared tools:

- ▶ PCA / SVD for initialization and factor extraction
- ▶ Grid search over nonlinear parameters
- ▶ Information criteria adapted for model complexity
- ▶ Monte Carlo validation before empirical application

Future Research

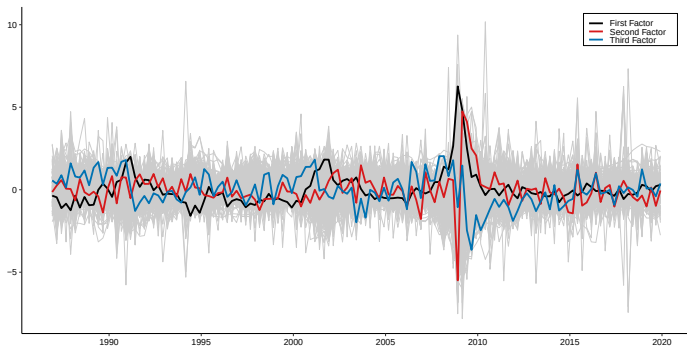
- ▶ **Asymptotic distribution theory** for the smooth transition and threshold factor estimators (Essays I and II): confidence intervals and formal hypothesis tests
- ▶ **Impulse response functions** under nonlinear factor VECMs: regime-conditional transmission of global shocks
- ▶ **Multiple regimes**: extending Essays I and II to $K > 2$ regimes with endogenous regime selection
- ▶ **Real-time financial cycle monitoring**: extending Essay III to near-real-time updating with mixed-frequency data
- ▶ **Cross-country spillovers**: embedding country-specific DFM indicators into a global model of systemic risk contagion

Thank You

Questions?

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Backup: STFM — Factor Estimates



Estimated factors from the STFM (black) vs. linear DFM (gray). Shaded areas: NBER recessions.

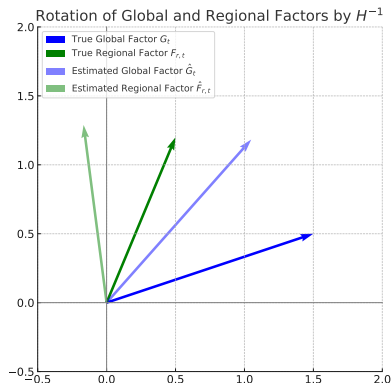
Backup: Financial Cycle — Variable Weights

Factor loadings $\hat{\lambda}_i$ by variable and country (approximate):

- ▶ **Highest loadings:** house price growth, credit-to-GDP growth, bank leverage
- ▶ **Moderate loadings:** equity prices, debt-service ratio
- ▶ **Lowest loadings:** current account (stabilising role — often countercyclical)

Loadings are **country-specific**: their relative importance varies with national financial structures.

Backup: Essay II — Rotation Diagram



Two-level factor structure: global factor (top) connected to group-specific factors (middle) and observables (bottom).